

Homework 6

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In the future, I will submit assignments in the form you requested in your recent email. I have typed this assignment because I am sick and unfortunately all I have in my room right now is my laptop.

(1) Let $\mathfrak{g}$ be the abelian Lie algebra of translations. We first verify that ω is skew-symmetric:

$$\omega((a, b), (c, d)) = ad - bc = -(cb - da) = -\omega((c, d), (a, b)).$$

Now we show $d\omega = 0$ for d the Chevalley-Eilenberg differential:

$$d\omega(u, v, w) = -\omega([u, v], w) + \omega([u, w], v) - \omega([v, w], u) = 0,$$

where we have used the fact that $\mathfrak{g}$ is abelian, meaning all brackets vanish. Thus ω is a 2-cocycle on $\mathfrak{g}$.

(2) The central extension determined by ω is the short exact sequence of Lie algebras

$$0 \longrightarrow \mathbb{R} \longrightarrow \mathfrak{g} \oplus \mathbb{R} \longrightarrow \mathfrak{g} \longrightarrow 0$$

with maps determined in the evident way. The Lie bracket on $\mathfrak{g} \oplus \mathbb{R}$ is, for $u, v \in \mathfrak{g}$ and $x, y \in \mathbb{R}$,

$$[(u, x), (v, y)]_{\mathfrak{g} \oplus \mathbb{R}} = ([u, v]_{\mathfrak{g}}, \omega(u, v)).$$

Since the Lie bracket in $\mathfrak{g}$ is trivial, we obtain that if $u = (a, b)$ and $v = (c, d)$

$$[(u, x), (v, y)]_{\mathfrak{g} \oplus \mathbb{R}} = (0, ad - bc).$$

(3) To show that that this is the Heisenberg Lie algebra, we identify

$$((a, b), x) \longleftrightarrow \begin{pmatrix} 0 & a & x \\ 0 & 0 & b \\ 0 & 0 & 0 \end{pmatrix}.$$

This map is evidently $\mathbb{R}$ -linear, so we need only show that this map respects the Lie bracket. Since the Lie bracket for the Heisenberg Lie algebra is fully determined by the basis elements $X = (1, 0, 0)$, $Y = (0, 1, 0)$, and $Z = (0, 0, 1)$ satisfying $[X, Y] = Z$, $[X, Z] = 0$, and $[Y, Z] = 0$, it suffices to show this holds in $\mathfrak{g} \oplus \mathbb{R}$.

$$[X, Y] = [((1, 0), 0), ((0, 1), 0)] = ((0, 0), 1 - 0) = ((0, 0), 1) = Z$$

$$[X, Z] = [((1, 0), 0), ((0, 0), 1)] = ((0, 0), 0) = 0$$

$$[Y, Z] = [((0, 1), 0), ((0, 0), 1)] = ((0, 0), 0) = 0$$

Therefore the central extension $\mathfrak{g} \oplus \mathbb{R}$ determined by ω is the Heisenberg Lie algebra.